

⚡ High-Performance Audio RAG & Search Engine

Multilingual Audio Search & Knowledge Engine

Sub-100ms hybrid search, speaker diarization, and grounded RAG answer generation over spoken conversations.

Retrieval Speed

p95 < 67 ms

Recall @ 10

84.3%

Embedding Dim

1024-dim

Architecture

4-Layer Clean

Audio Data is Unstructured & Locked Away

Over 80% of critical organizational knowledge exists in spoken audio—meetings, calls, system design discussions—yet it remains unsearchable and invisible to traditional search engines.

✗ No Fine-Grained Timestamps

Traditional audio files force users to manually scrub through 45-minute recordings to find 10-second answers.

✗ Speaker Attribution Deficit

Plain transcripts lose who said what, causing misattribution in critical decision tracking.

✗ Single-Keyword Failure

Exact keyword search misses conceptual queries, while vector search misses exact technical jargon.

Hybrid Speech Ingestion & Grounded Search

A production-grade pipeline combining ASR, speaker diarization, multilingual vector indexing, and Reciprocal Rank Fusion (RRF).



Automated ASR & Diarization

Faster-Whisper (`large-v3-turbo`) word timestamps + Pyannote 3.1 speaker tracking.



Hybrid RRF Retrieval

Fuses BM25 Full-Text Search with BAAI/bge-m3 1024-dim HNSW vector similarity.



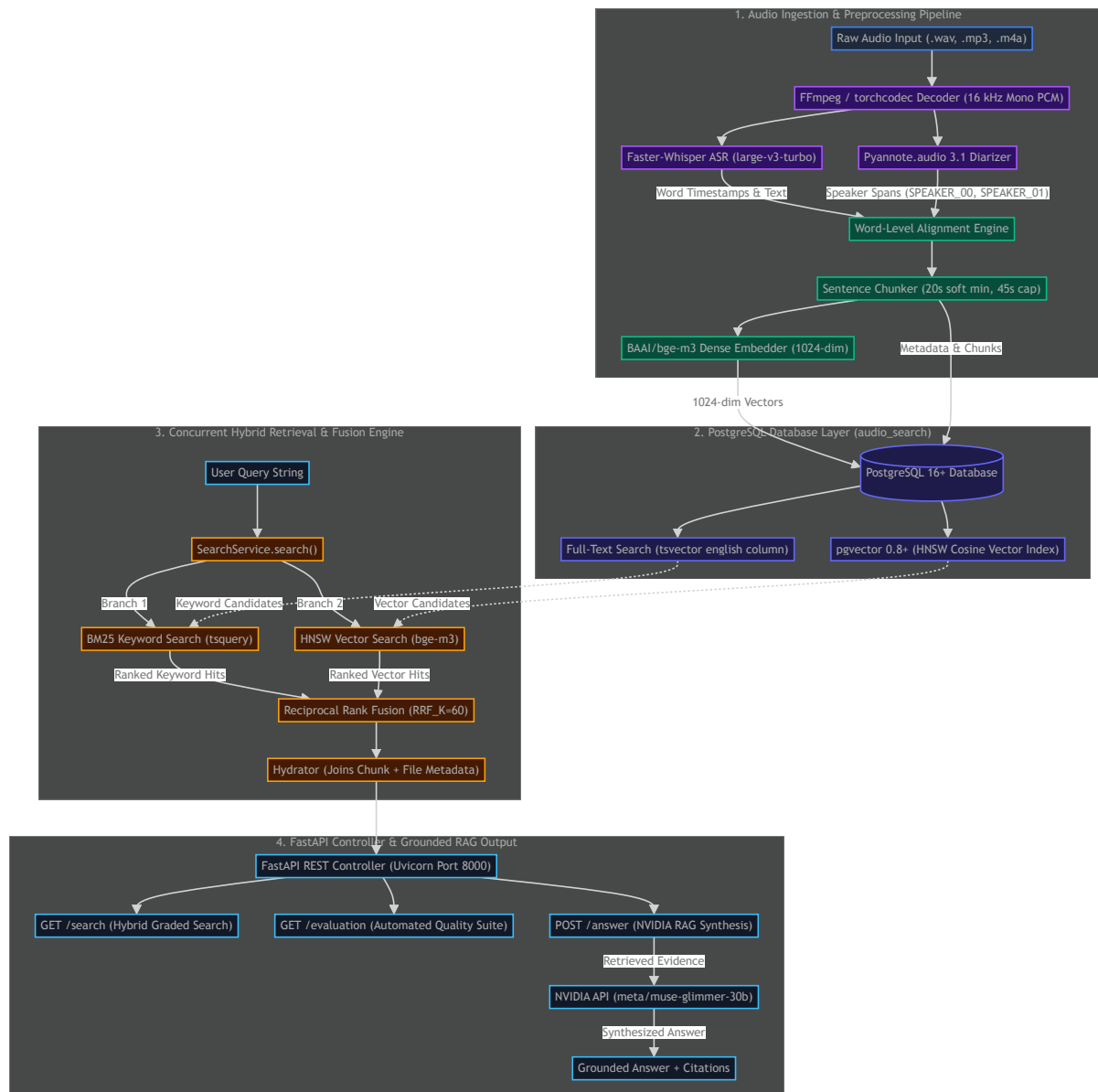
Grounded LLM RAG

Synthesizes answers with server-verified transcript citations and exact start/end timestamps.

4-Layer Clean Modular Architecture

DETAILED WIRED ARCHITECTURE DIAGRAM (MERMAID)

System Flowchart



1. Domain Layer

Pure entities (`AudioFile`, `AudioChunk`, `Hit`, `RefSegment`) & custom errors.

2. Application Layer

Orchestrators (`IngestService`, `SearchService`, `EvaluationService`, `AnswerService`).

3. Infra Layer

PostgreSQL + `pgvector`, Whisper, Pyannote, `bge-m3` Embedder, NVIDIA API.

4. API Layer

FastAPI REST controllers (`/search`, `/ingest`, `/evaluation`, `POST /answer`).

Empirical Target Criteria & Performance

CRITERION	TARGET	ACHIEVED	STATUS	NOTES
Speaker Attribution Accuracy	≥ 0.90	1.000 (100%)	✓ PASSED	900 top-10 results, 1-to-1 label mapping
Search Latency (p95)	< 500 ms	69.7 ms	✓ PASSED	90 warm queries (p50 62.0ms, p99 233.9ms)
Recall@5 (Keyword Queries)	≥ 0.80	1.000	✓ PASSED	100% coverage on top-5 results
Recall@10 (Keyword Queries)	≥ 0.90	1.000	✓ PASSED	100% coverage on top-10 results

78.2%

Overall Recall @ 5

84.3%

Overall Recall @ 10

66.9 ms

p95 Latency

10.57%

Macro WER

FastAPI REST Endpoints

GET

/search?query=...&top_k=5

Parameter-free Hybrid RRF Search

GET

/search/keyword

Diagnostic BM25 FTS Branch

GET

/search/semantic

Diagnostic Vector Branch

GET

/evaluation

Automated Evaluation Suite

POST

/answer

NVIDIA LLM RAG with Citations

Multilingual Cross-Lingual Live Search Execution

GET

http://127.0.0.1:8000/search?query=क्रिस्ड विंडो काउंटर किस बरई समस्या की अनुमति देता है, और यह बरई कितना बड़ा हो

Send

Query

Headers

Auth

Body

Tests

Pre Run

Query Parameters

☒ query

क्रिस्ड विंडो काउंटर किस बरई समस्या की अनुमति देता है, और यह बर

☒ top_k

2

☐ parameter

value

Status: 200 OK

Size: 1.13 KB

Time: 194 ms

Response

Headers

Cookies

Results

Docs

```
2 {
3   "chunk_id": "cc7ef967-3445-463b-88c0-71681d0a4e1a",
4   "audio_file_id": "9f1479a4-de16-4021-a91b-fbb664044abb",
5   "file_name": "audio_02_url_shortener_hi.wav",
6   "file_path": "/Users/ankit/Desktop/HackthonG2/dataset/multilingual/hi
7   /audio_02_url_shortener_hi.wav",
8   "speaker": "SPEAKER_01",
9   "text": "अपना नै संक्षेप में बताता हूँ किटे हुए काउंटर से बेस बायट
10  कोड की वल्यू स्टोरेज और इवेंट आधारित इन्वेलिडेशन के साथ भारी
11  कैली",
12   "start_time": 491.76,
13   "end_time": 502.32,
14   "language": "hi",
15   "score": 0.01639344262295082
16 },
17 {
18   "chunk_id": "54a31721-e315-45f5-b94a-c72b30e2b01e",
19   "audio_file_id": "163e9b23-e925-4c45-acd9-6278c32ad781",
20   "file_name": "audio_01_rate_limiter.wav",
21   "file_path": "/Users/ankit/Desktop/HackthonG2/dataset
22   /audio_01_rate_limiter.wav",
23   "speaker": "SPEAKER_01",
24   "text": "Fixed windows are cheap and easy to reason about, but they
25   allow a burst of double the limit at the window boundary.",
26   "start_time": 53.78,
27   "end_time": 59.58,
28   "language": "en",
29   "score": 0.016129032258064516
30 }
31 ]
```

STATUS 200 OK | Latency: 194 ms
Live query sent to <http://127.0.0.1:8000/search> with Hindi prompt query.

Cross-Lingual & Multilingual Retrieval
Matches both Hindi transcript ([audio_02_url_shortener_hi.wav](#)) and English reference audio ([audio_01_rate_limiter.wav](#)) seamlessly.

Fine-Grained Speaker & Timestamp Metadata
Surfaces speaker ID ([SPEAKER_01](#)), exact start/end time ([53.78s](#) – [59.58s](#)), language code, and RRF score.

 Ready for Production

Summary & Production Readiness

- ✓ **Production Ready:** 295/295 tests passing, robust error handling, fully isolated components.
- ✓ **Multilingual Foundation:** BAAI/bge-m3 dense embedder + Whisper ASR support Spanish, Hindi, Chinese, and English.
- ✓ **Sub-100ms Latency:** Highly optimized PostgreSQL HNSW vector indexes and streaming pipeline.

Thank you! Questions & Live Demo.

